

Python Fundamentals — Practice Questions

Topics: Data Type & Variable | Operator & Expression | Conditional Statement | Loop & Iterations | Function | Slicing

Instructions: Write your code / answer for each question. Test your code in a Python IDE or Jupyter Notebook. Try to solve each question without looking at the reference notebook first.

Q1. [Data Type & Variable]

Create variables to store your name (string), age (integer), height in meters (float), and whether you are a student (boolean). Print each variable along with its data type using `type()`.

Q2. [Data Type & Variable]

Take a number as a string, e.g. `num_str = "42"`. Convert it to an integer, then to a float, and finally back to a string. Print the value and `type()` at every step.

Q3. [Operator & Expression]

Given two numbers `a = 18` and `b = 5`, write expressions to find and print their sum, difference, product, quotient, floor division, remainder, and `a` raised to the power of `b`.

Q4. [Operator & Expression]

Given `x = 12` and `y = 7`, write a single expression using logical operators (and, or, not) to check whether at least one of them is greater than 10, AND both of them are positive numbers. Print the result.

Q5. [Conditional Statement]

Write a program that takes a person's age as input and prints whether they are a 'Child' (0-12), 'Teenager' (13-19), 'Adult' (20-59), or 'Senior Citizen' (60+) using `if-elif-else`.

Q6. [Conditional Statement]

Write a program that checks whether a given year is a leap year. (A year is a leap year if it is divisible by 4, but century years must also be divisible by 400.)

Q7. [Loop and Iterations]

Using a `for` loop and `range()`, print all even numbers between 1 and 50. Then, using a `while` loop, calculate and print the sum of all these even numbers.

Q8. [Loop and Iterations]

Write a program using a `for` loop with `break` and `continue`: print numbers from 1 to 20, skip multiples of 3 using `continue`, and stop the loop completely once the number 18 is reached using `break`.

Q9. [Function]

Write a function named `is_prime(n)` that returns `True` if a number is prime and `False` otherwise. Use it inside a loop to print all prime numbers between 1 and 30.

Q10. [Function + Slicing]

Write a function named `reverse_and_slice(text)` that takes a string, returns the reversed string, and also returns the first half of the original string using slicing (no built-in reverse methods other than slicing allowed).

Hints / Key Concepts to Use

Q.No	Topic	Key Concept / Function to Use
1	Data Type & Variable	str, int, float, bool, type()
2	Data Type & Variable	int(), float(), str() type casting
3	Operator & Expression	+ - * / // % **
4	Operator & Expression	and, or, not, comparison operators
5	Conditional Statement	if / elif / else
6	Conditional Statement	% (modulus) with nested if / and, or
7	Loop and Iterations	for with range(start, stop, step), while
8	Loop and Iterations	for, if, continue, break
9	Function	def, return, % modulus, loop inside range check
10	Function + Slicing	def, [::-1] for reverse, [:len(text)//2] for first half

Tip: Try to write and run each solution yourself in a Jupyter Notebook or Python IDE before checking any reference solution. Good luck!